

UTM FOUNDATION
2019/2020-3

FSPP0034 PHYSICS II

TEST 2

10 March 2020

ANSWER ALL THREE QUESTIONS

DURATION: 1 Hour

NAME:.....

IC NO:.....

LECTURER:.....

SECTION:.....

QUESTION 1 (10 MARKS)

- (a) i. In the relation $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ which pairs of the vectors ($\mathbf{F}$, $\mathbf{v}$, $\mathbf{B}$) are always at 90° ? Which can be at other angles?

F is always perpendicular to both v and B. v and B can be at any angle with respect to each other

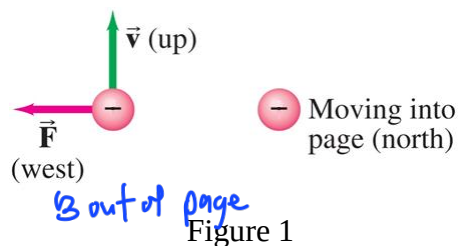
- ii. Write the equation of Ampère's law and explain the meaning.

$$B = \frac{\mu_0 I}{2\pi r}$$

- the magnetic field around a closed loop to the total current flowing through the loop

(2 marks)

- (b) Figure 1 shows a magnetic field exerts a force of 1.6×10^{-13} N toward the west on an electron moving vertically upward at a speed of 5.0×10^6 m/s. When moving horizontally in a northerly direction, the force on the electron is zero.



Determine:

- i. The direction of the magnetic field in this region.

Since the force is zero when the electron is moving north, the field must point in the north-south direction.

In order for the force to be to the west when the electron is moving up, the field must point south.

- ii. The magnitude of the magnetic field. **ANS: 0.2 T**

$$F = qvB$$

$$1.6 \times 10^{-13} = 1.67 \times 10^{-19} (5 \times 10^6) (B)$$

$$B = 0.192 \text{ T}$$

(4 marks)

- (c) Figure 2 shows a long straight cylindrical wire conductor of radius R carries a current I of uniform current density in the conductor.

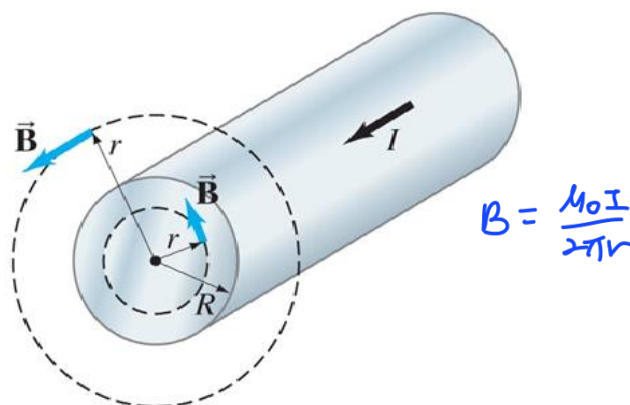


Figure 2

Determine the magnetic field due to this current at:

- i. Points outside the conductor ($r > R$)

$$r > R$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc}$$

$$B 2\pi r = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

ii. Points inside the conductor ($r < R$).

$$\begin{aligned}
 r &< R \\
 J &= \frac{I}{A} \\
 \frac{I}{A} &= \frac{I_{enc}}{A} \\
 \frac{I}{\pi R^2} &= \frac{I_{enc}}{\pi r^2} \\
 I_{enc} &= I \frac{\pi r^2}{\pi R^2} \\
 B(2\pi r) &= \mu_0 I_{enc} \\
 B &= \frac{\mu_0 I r}{2\pi R^2}
 \end{aligned}$$

iii. If $R = 2.0$ mm and $I = 30$ A, what is B at $r = 3.0$ mm? **ANS : 2.0×10^{-3} T**

$$\begin{aligned}
 r > R, \quad B &= \frac{\mu_0 I}{2\pi r} \\
 &= \frac{4\pi \times 10^{-7} (30)}{2\pi \times 3 \times 10^{-3}} \\
 &= 2 \times 10^{-3} \text{ T}
 \end{aligned}$$

(4 marks)

QUESTION 2 (10 MARKS)

Figure 3 shows a 1000-loop hexagonal coil of wire with the side $a = 2$ cm total resistance of $R = 50 \Omega$ is positioned perpendicular to a uniform 10 T magnetic field. The coil is quickly pulled (perpendicular to B) at a constant speed to a region where B is zero. At $t = 0$ s, the corner of the coil is at the edge of the field. It takes 0.50 s for the whole coil to reach the field-free region.

$$A = \frac{3\sqrt{3}a^2}{2} = 6\sqrt{3} \text{ cm}^2$$

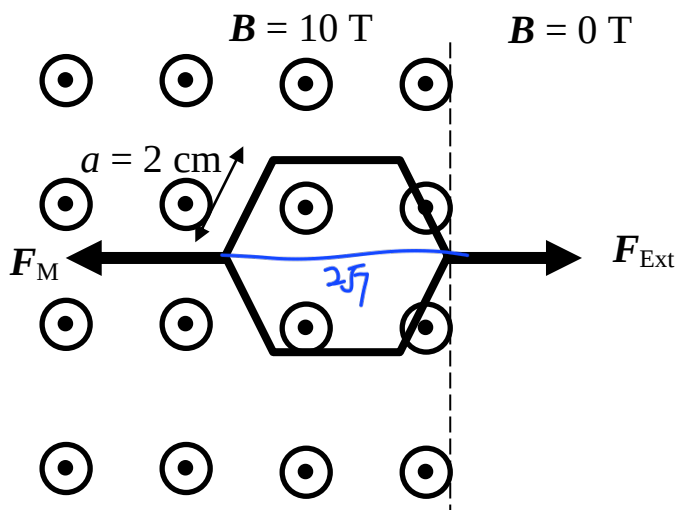


Figure 3

Determine:

- i. The rate of change of flux through the coil **ANS : 0.02 Wb/s**

$$\begin{aligned}\frac{d\Phi}{dt} &= \frac{BA \cos \theta}{t} = \frac{\Delta BA \cos \theta}{t} = \frac{(-10.0)(\sqrt{3} \times 10^{-4}) \cos 90^\circ}{0.50} \\ &= -0.021 \text{ Tm}^2\text{s}^{-1}\end{aligned}$$

- ii. The emf induced, emf **ans : 20 V**

$$\begin{aligned}\mathcal{E} &= -N \frac{d\Phi}{dt} \\ &= -1000(-0.021) \\ &= 21 \text{ V}\end{aligned}$$

- iii. The current induced, I **ans : 0.4 A**

$$\begin{aligned}\mathcal{E} &= IR \\ 21 &= I(50) \\ I &= 0.42 \text{ A}\end{aligned}$$

- iv. The energy dissipated in the coil, E **ans : 4 J**

$$\begin{aligned}E &= Pt = I^2 R t \\ &= (0.42)^2 (50) (0.50) \\ &= 4.41 \text{ J}\end{aligned}$$

- v. The external force, F_{ext} ans : 10 N

$$F_{\text{ext}} = I l B \sin \theta$$
$$= 0.42(27)(10) \sin$$

(10 marks)

QUESTION 3 (10 MARKS)

- (a) Define and state equations for the two types of inductance.

Self inductance : the process of producing an inducing emf in the coil due to change of current of current flowing through the same coil

$$L = \frac{\Phi}{I}, L = \frac{\mu_0 N^2 A}{l}$$

Mutual Inductance : the process of producing an induced emf in one coil due to the change of current in another coil

$$M = \frac{N_2 \Phi_2}{I_1}$$

(2 marks)

- (b) An air-filled cylindrical inductor has 2,600 turns, and it is 2.5 cm in diameter and 28.2 cm long.

- i. Calculate its inductance. ans : 0.015 H

$$L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} (2600^2) (\pi \times 1.25 \times 10^{-2}^2)}{0.282}$$
$$= 0.0148 \text{ H}$$

- ii. How many turns would you need to generate the same inductance if the core were iron-filled instead? **ans : 75 turns**

Assume the magnetic permeability of iron is about 1,200 times that of free space.

$$0.0148 = \frac{4\pi \times 10^{-7} \times 1200 \times N^2 \times (1.25 \times 10^{-2})^2 \pi}{0.282}$$

$$N^2 = 5636.86$$

$$N = 75.08 \text{ turns}$$

$$\approx 75 \text{ turns}$$

(3 marks)

- (c) At $t = 0$, the current through a 45.0 mH inductor is 50.0 mA and is increasing at the rate of 115 mA/s.

- i. What is the initial energy stored in the inductor? **ans : $5.63 \times 10^{-5} \text{ J}$**

$$\begin{aligned} U &= \frac{1}{2} LI^2 \\ &= \frac{1}{2} (45 \times 10^{-3}) (50 \times 10^{-3})^2 \\ &= 5.625 \times 10^{-5} \text{ J} \end{aligned}$$

- ii. How long does it take for the energy to increase by a factor of 5.0 from the initial value? **ans : 0.538 s**

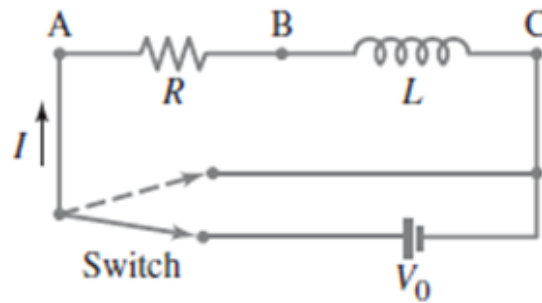
$$\begin{aligned} U \times 5 &= 2.8125 \times 10^{-4} \text{ J} \\ 2.8125 \times 10^{-4} \text{ J} &= \frac{1}{2} L \left(I + \frac{dI}{dt}(t) \right)^2 \\ 2.8125 \times 10^{-4} &= \frac{1}{2} (45 \times 10^{-3}) (50 \times 10^{-3} + 115 \times 10^{-3} t)^2 \\ t_1 &= -1.407, t_2 = 0.537 \text{ s} \end{aligned}$$

(3 marks)

- (d) How many time constants does it take for the potential difference across the resistor in an LR circuit like in Figure 4 to drop to 2.5 % of its original value, after the switch is moved to the upper position, removing V_0 from the circuit? ans :3.7s

$$V_1 = x$$

$$V_2 = 0.025x$$



$$\tau = \frac{L}{R}$$

Figure 4

$$V = V_0 e^{-\frac{t}{\tau}}$$

$$V = 0.025 V_0$$

$$0.025 = e^{-x},$$

$$-x = \ln(0.025)$$

$$-x = -3.69$$

$$x = 3.69$$

$$t = 3.69\tau$$

$$e^{-\frac{t}{3.69}} = 0.025$$

$$\ln(0.025)$$

$$\frac{t}{3.69} = 3.69$$

(2 marks)

APPENDIX

List of constant

$$e = -1.60 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$g = 9.80 \text{ m s}^{-2}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

$$\text{Coulomb's constant, } k = 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

List of formula

$$F = q v B \sin \theta$$

$$B = \frac{\mu_0 I}{2\pi d}$$

$$F = ILB \sin \theta$$

$$U = \frac{1}{2} LI^2$$

$$P = IV$$

$$L = \frac{\mu_0 N^2 A}{l}$$

$$B = \mu_0 n I$$

$$I = I_{\max} (1 - e^{-t/\tau}) = \frac{V}{R} (1 - e^{-t/\tau})$$

$$\varepsilon = -L \frac{dI}{dt}$$

$$\tau = \frac{L}{R}$$

$$I = \frac{\varepsilon}{R}$$

$$\varepsilon = NB\omega A$$

$$\varepsilon = Blv$$

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

$$\Phi_B = BA \cos \theta$$

$$M_{21} = \frac{N_2 \Phi_{21}}{I_1}$$

$$M_{21} = \mu_0 \frac{N_1 N_2}{l} A$$

$$\mathcal{E}_A = -M \frac{dI_B}{dt}$$

$$L \frac{dI}{dt} + RI = V_0$$

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

$$\text{Area of hexagon} = \frac{3\sqrt{3} (a^2)}{2}$$